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Federated Class-Incremental Learning with New-Class Augmented Self-Distillation

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Research Objectives

This study aims to address the catastrophic forgetting problem caused by dynamic category expansion in federated learning. The authors propose a novel federated class-incremental learning method, FedCLASS, which enhances the missing new category information in historical models to achieve more precise knowledge transfer, thereby improving the overall performance and stability of models in multi-task, dynamic data environments.

Research Method

FedCLASS introduces a new-class augmented self-distillation mechanism, which integrates the current model’s predictions for new categories with the historical model’s predictions for old categories to reconstruct a complete class probability distribution. The specific methods include:

- Knowledge modeling based on conditional probabilities to separately handle scores for old and new categories;
- Designing a new class-augmented loss function that balances cross-entropy and Kullback–Leibler divergence;
- Theoretically proving the rationality and reliability of FedCLASS.

Research Results

Experiments demonstrate that FedCLASS outperforms existing baseline methods in terms of both global accuracy and average forgetting rate across four datasets and under various task incremental settings.

Research Conclusions

FedCLASS effectively mitigates the catastrophic forgetting problem in federated class-incremental learning. It is the first to establish a theoretical probabilistic modeling framework combining incremental new categories with historical knowledge. Through new-class augmented self-distillation, FedCLASS achieves more complete knowledge transfer, significantly improving the model's global performance.